SANYAM JHA

+91-8866673751 | sanyam.jha.cd.ece20@iitbhu.ac.in | GitHub | LinkedIn

PROFILE SUMMARY

Application Engineer with a B.Tech. in Electronics and experience in machine learning, natural language processing, computer vision, reinforcement learning, federated learning, and AI-based product data workflows. Experienced in developing LSTM-based models for NER, embedding-based semantic automapping, Vision Transformers for infrared imaging, and reinforcement learning approaches for Wireless Sensor Networks. Worked with large-scale text processing, distributed training, transfer learning, and full-stack application development. Academic project experience includes UAV-IoT networks, federated learning, and early breast cancer detection using infrared imaging.

EDUCATION

Indian Institute of Technology (Banaras Hindu University), Varanasi — Aug 2020–May 2024
Bachelor of Technology in Electronics — CGPA: 9.08/10.0

WORK EXPERIENCE

***Application Engineer*, SCM PIM Team, Oracle, Hyderabad** — Jun 2024 – Present

- Built a custom vector database using FAISS to add filtered similarity search, a capability missing from Oracle AI Agent Studio's built-in vector DB, enabling an embedding-based semantic automapping workflow for product attributes that replaced LLM reasoning and cut latency from ~120s to ~20s.
- Worked within the team responsible for developing and maintaining core services underlying the Product Management module, used by teams across the org and by customers running large product catalogs; used a Java-based ADF framework to build REST services for business logic implementation using JavaScript-based Redwood UI, modernizing legacy SOAP integrations in Fusion SCM Product Management.
- Used Oracle Visual Builder Studio (VBCS) to rebuild 4 ADF-based UI workflows into Redwood UI, moving them from SOAP to REST and aligning them with current industry UI development standards.
- Diagnosed and fixed recurring failures in the team's LRG test suites (automated regression tests covering service reliability under predictable requests) that had been failing due to DB seeding issues; identified and resolved the root cause in 3 services, restoring stable, reliable test runs. Also used PL/SQL, JUnit, and Selenium for data model updates and automated validation of mapping quality using Top-K semantic evaluation.

INTERNSHIP EXPERIENCE

***Application Engineer Intern* , Oracle, Hyderabad** — May 2023 – Jun 2023

- Built and trained LSTM-based ML models using TensorFlow, Pandas, and Keras to solve NER-type problems on out-of-vocabulary (OOV) industry data, such as shorthand codes like "sc 4 br" (screw, 4 inch, bronze), where different vendors used different codes and IDs for functionally identical parts; used LLMs to generate standardized text output for this OOV data.
- Used Apache Spark and Spring Boot to build a text ingestion pipeline processing at least 4,000,000 records daily.
- Built a POC for a vendor-agnostic standardized part-profiling feature in Fusion Applications, addressing the problem of hospitals sourcing identical surgical parts from multiple vendors with inconsistent part representations; the POC focused on demonstrating feasibility rather than performance.
- Presented the POC to executives, and it became a core part of the service the PM team was building, with plans to extend it into end-to-end inventory automation, including auto-ordering of exhausted parts.

***Web Development Intern,* Saarthi.ai, Bengaluru** — Dec 2021 – Feb 2022

- Built a MERN stack dashboard for Saarthi.ai, a voice AI SaaS company building Indic-language voice agents to replace call center workers for banks and financial institutions calling customers about loans and other financial services; the dashboard let the business track usage of its voice AI agents state-by-state across India, using a wireframe based on the map of India as the core interface.
- Developed the CRUD APIs and integrated them with the back-end, targeting faster rendering of this real-time usage data.
- Developed the API's authentication layer using JWT and added email notifications using SendGrid.

PROJECTS

Federated Learning enabled Small World (SW) UAV-IoT Network using PPO, Department of ECE, IIT BHU — Jan 2023 – Nov 2023

- Developed a novel RL-based approach introducing small-world characteristics in Wireless Sensor Networks, achieving 51% lower APL and 117% higher Small-Worldness.
- Trained a PPO-based RL agent to improve network robustness independent of the underlying message-passing protocol, using LEACH, A*, Dijkstra, and multi-hop as reference protocols (consistent with prior published work) to gather message-passing data, and evaluated robustness across energy consumption, throughput, and latency.
- Simulated distributed training with the Flower framework on ResNet-101 using Food-11 and MNIST, observing only a 0.11% accuracy drop compared to centralised ML.
- Demonstrated that the federated learning architecture remained efficient under skewed data distributions by comparing precision, recall, and F1-score against centralised image classification models, observing a positive delta in these metrics under skew.

Vision Transformers & Infrared Imaging for Early Cancer Detection, Department of CSE, IIT BHU — Dec 2023 – Mar 2024

- Used Vision Transformers and infrared imaging to develop a novel method for early breast cancer detection.

- Used Transfer Learning to fine-tune Google's vit-base-patch16-224-in21k on the DMR-IR Dataset; this project deepened understanding of Vision Transformer architecture, including its application to image data beyond the text domain it was originally built for, and of infrared imaging as a technique, building on prior coursework in hyperspectral imaging.
- Achieved test accuracy of 96.8% and F1 score of 96.7%.

PROGRAMMING SKILLS

- **Languages:** Python, JavaScript, C++, PL/SQL, Java, Go
- **Technologies:** PyTorch, TensorFlow, NumPy, Spark, Pandas, Transfer Learning, Reinforcement Learning, Computer Vision, Natural Language Processing, React